

Analog Lab Report Week 5

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Objective :

This experiment is a continuation of Week 4 Experiment. Here, we are adding two waves which we will be getting from Vout of Band Pass Filter 1 and 2.

List of Components :

1. 1 – MCP6004
2. Capacitors and resistors
3. Power Supply (ADALM 1000)
4. Breadboard

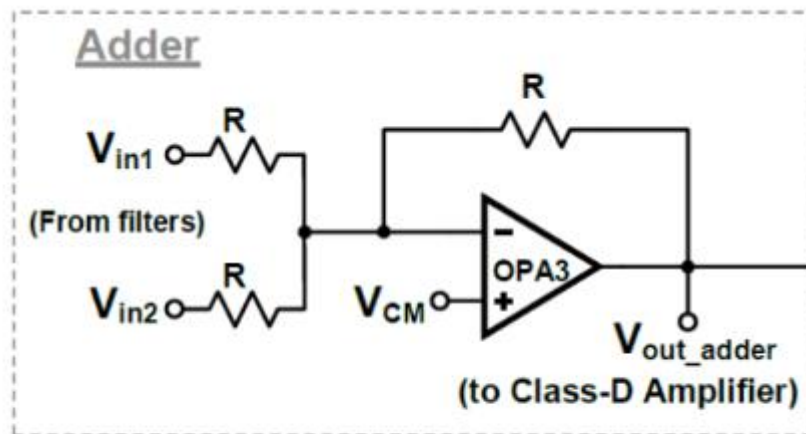


Figure 9 Schematic diagram of an adder.

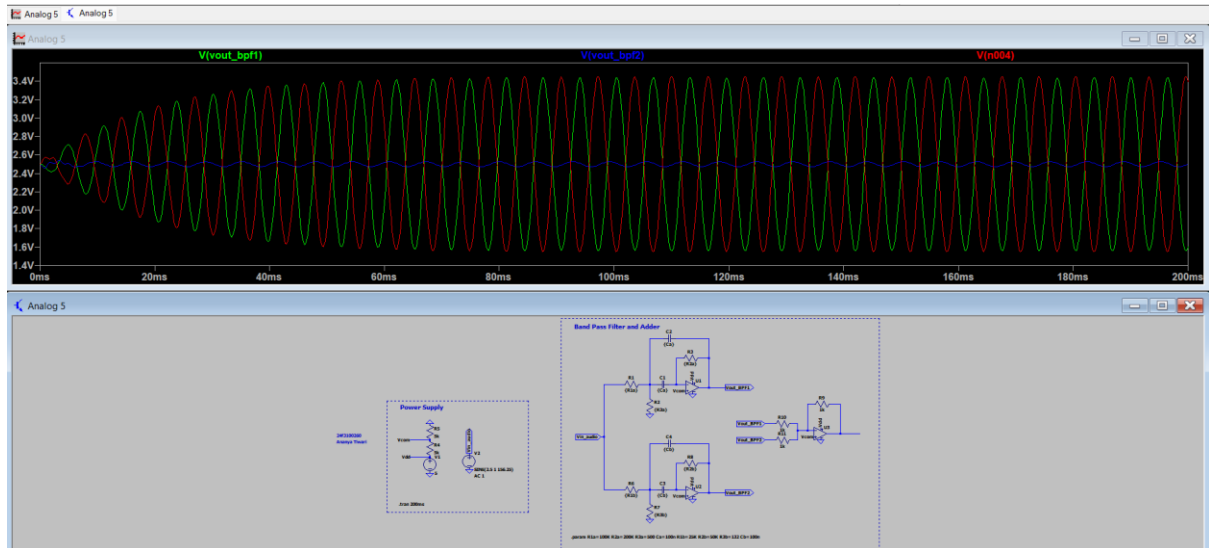
LTspice Schematics and Output :

Adder Vout while keeping the input frequency of sin wave as 156.25 Hz.

Green signal – Vout of BPF 1

Blue signal – Vout of BPF 2

Red signal – Adder Output

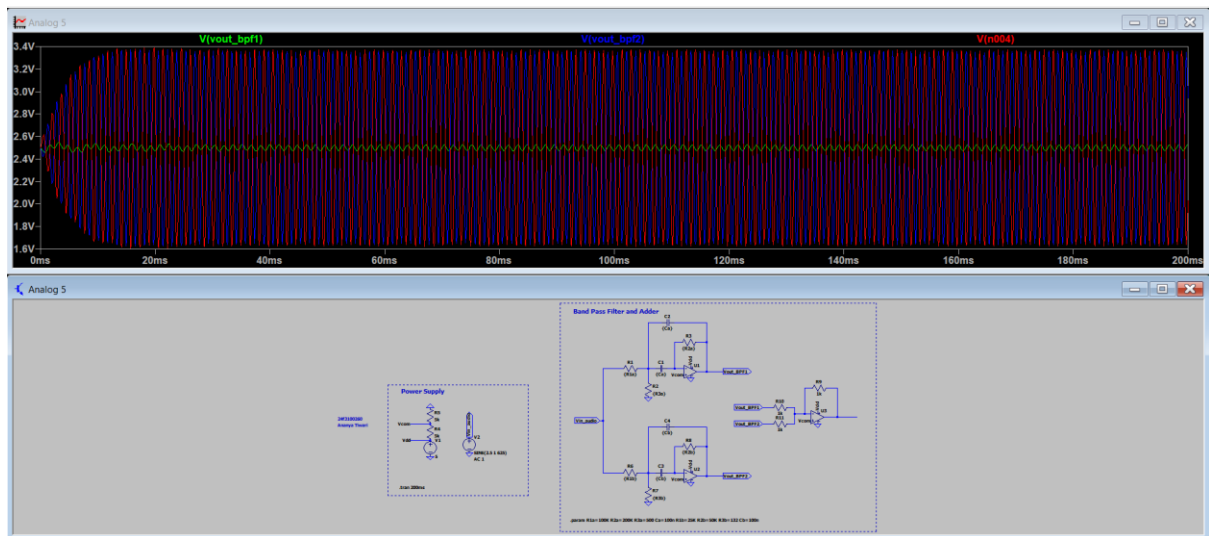


Adder Vout while keeping the input frequency of sin wave as 625 Hz.

Green signal – Vout of BPF 1

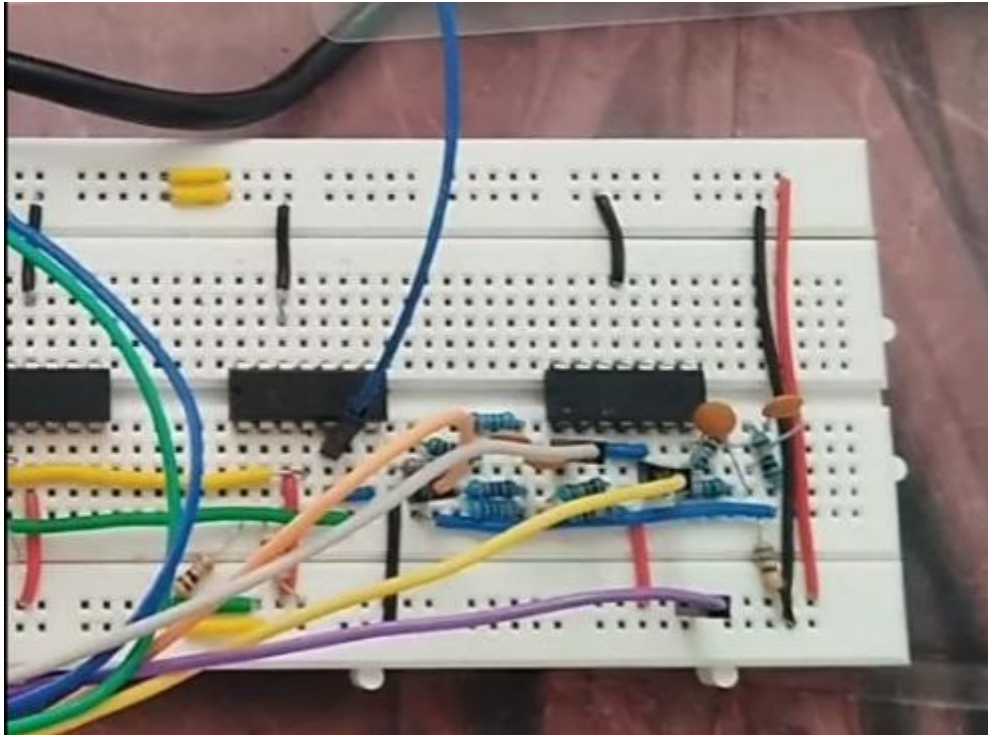
Blue signal – Vout of BPF 2

Red signal – Adder Output



Physical Circuit :

The circuit configuration remains identical to Week 4, except a different operational amplifier within the same IC is utilized for the adder. Consequently, the adder output will be measured from a different pin.



Physical Circuit Wire Connections :-

Green wire and Blue wire – GND

Yellow – 5V

Orange wire – Channel B (*input*)

White wire – Channel A (*output*)

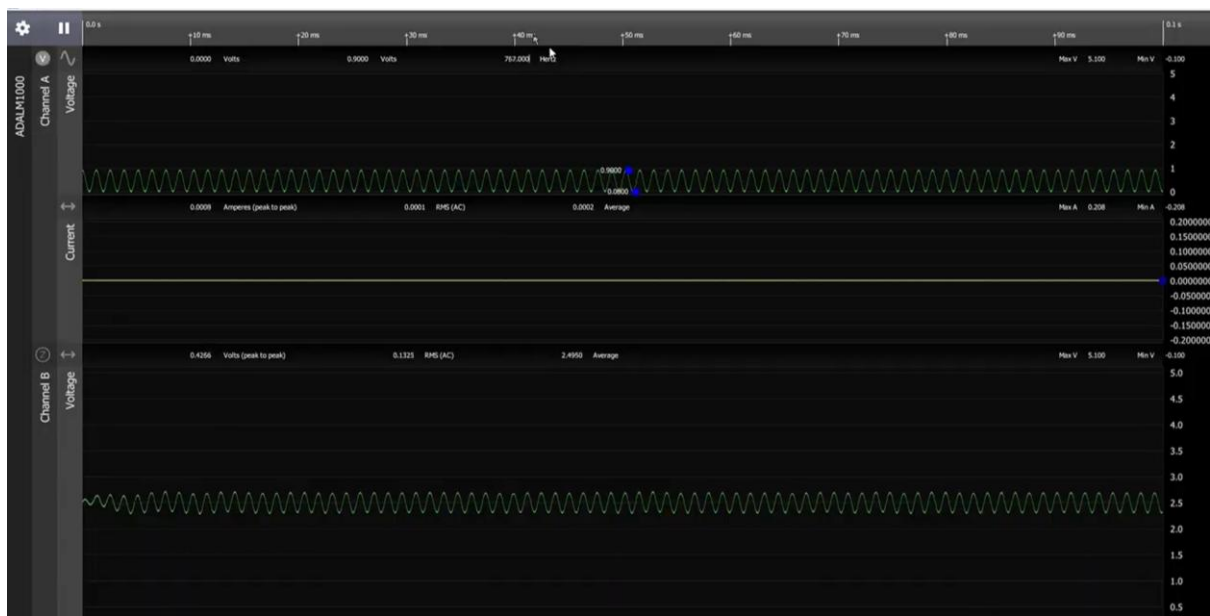
Purple wire – 2.5V

Physical Circuit Output :-



First one around frequency of 156.25 Hz.

Second one around frequency of 625 Hz.



Conclusion :-

The experiment was successful, with the physical hardware output validating the simulation models.